

## Quantum Approach to Solids - Problem Sheet 4

Dr. Andrew J. Morris

1. **Classical vs quantum description of metals** Calculate the classical mean kinetic energy for free electrons in sodium at room temperature. Use this to calculate the de Broglie wavelength of the electrons. Compare this to the mean separation  $d$  of the atoms. Comment on this.
2. **Electrons in a Fermi gas** In the free electron model, the electrons may be described as a Fermi gas.
  - (a) What principle must the electrons obey? State this principle, briefly.
  - (b) What does the Fermi energy,  $E_F$ , represent?
  - (c) For an electron to participate in electrical conduction, what can you state about its energy?
3. **A 1-D conductor** A 1-D metal is made into a wire consisting of 10,000 atoms. The equilibrium atomic separation is 3 Å. Each atom contributes one electron to the Fermi gas. Calculate the Fermi energy for this material.
4. **Nearly free electron model** Draw a labelled sketch of the dispersion relation for the electrons in a 1-D metal, with atomic separation  $a$ , in the nearly free electron model. Explain the deviations from the dispersion relation for the free electron model.